

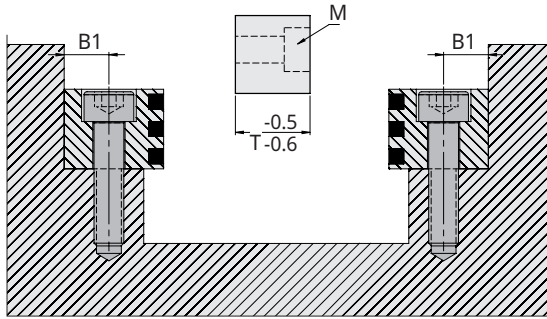
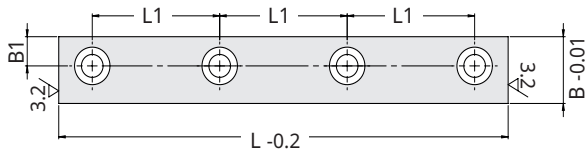
NEW



Wear Plate for Core Side

Self-Lubricating - Heat Resistance: 150°C

Code: **G172**



B	T	L	L1	B1	Hole	M
20	12	80	60	9	2	M6
		125	52.5		3	
	16	100	40		4	
		160	70		3	
25	22	125	50.5	11	4	M8
		200	58		3	
	30	160	68		4	
		250	74		4	
32	38	160	64	14	3	M10
		250	72		4	



Order:
G172. B x L

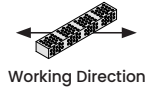
Material:

Bronze + graphite inserts



Download
www.guvenal.net

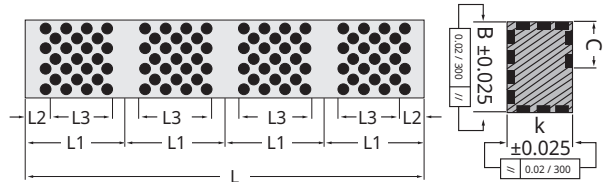
3D CAD - Formats:
CATIA, STEP, IGES 3D, X_T



Wear Plate Two-Way Sliding Type

Self-Lubricating - Heat Resistance: 150°C

Code: **G123**



k	B	L	C	L1	L2	L3
10.3	15.3	75	6	25	3	6
		100				
		150				
		200				
		250				
		300				
15.3	25.3	105	8	35	4	8
		140				
		175				
		210				
		245				
		280				
		315				
		350				
		385				
		420				
25.3	35.3	135	12	45	5	10
		180				
		225				
		270				
		315				
		360				
		405				
		450				
		496				
35.3	45.3	165	16	55	6	12
		220				
		275				
		330				
		385				
		440				
		495				



Order:
G123. k x L x B

Material:

Bronze + graphite inserts



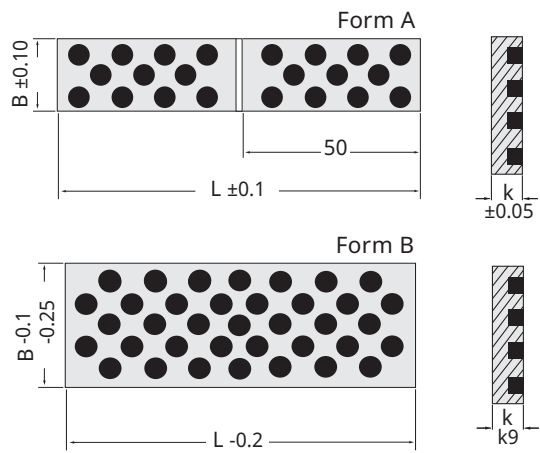
Die & Mould producers and users can be used by ordering standard product produced in 'L' length and cutting pieces suitable to their needs.

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Guiding Components



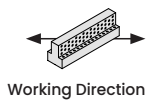
Wear Plate, Plain Type Code: **G122**
Self-Lubricating - Heat Resistance: 150°C



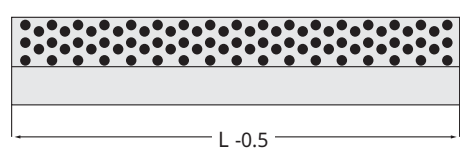
k	L	B	Form
5.3	302	20	A
		35	
		50	
10.3	302	20	A
		35	
		50	
5	50	25	B
	71		
	90		
5	50	40	B
	71		
	80		
	90		
6	100	40	B
	125	63	
	160		
	200		

Order: **G122.** k x L x B x Form

Long strip wear plates is compatible for use by perforated connection holes and cutting by user (in desired length). Working surfaces should be lubricated for once.

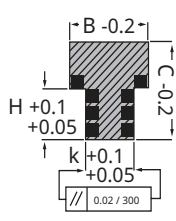


Wear Plate "T" Shaped Code: **G121**
Self-Lubricating - Heat Resistance: 150°C

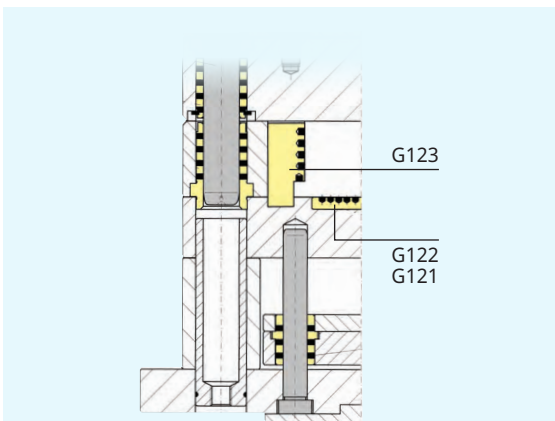


All wear plates friction surfaces are ground and all corners of wear plates are chamfered.

Order: **G121.** C x L



C	L	B	k	H
12	350	18	8	5
25	350	22	12	15
35	350	28	18	20



Usage of self-lub. wear plates on injection moulds:

Self-lubricating components provides load carrying capacity beyond expectations in low sliding speeds and a wide temperature range. The graphite inserts are positioned with an appropriate geometric structure. By this means maximum lubricating effect is achieved during sliding. They especially work better with hardened and ground bearings. Sliding surfaces should be slightly lubricated before starting to work.

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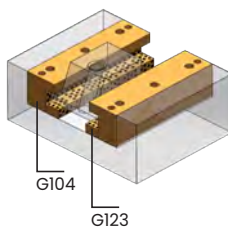
17

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Guiding Components



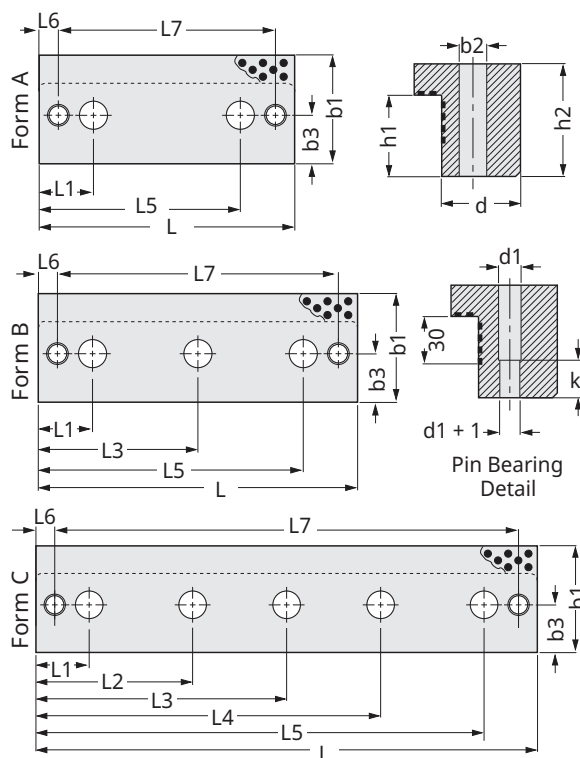
Mounting example



Wear Plate, "L" Shaped

Code: **G104**

Self-Lubricating & Two-Way Sliding Type



Form	b1	h1	b2	b3	d	d1	k	h2
A	55	55	37	20	13.5	10	23	39
A	70	75	50	30	17.5	12	35	55
A								
B								
C	85	90	63	38	22	16	45	65
A								
B								
C								

Form	L	L1	L2	L3	L4	L5	L6	L7
A	100	27.5	-	-	-	72.5	10	80
A	160					132.5		140
A	160					125		135
A	200	35	-	-	-	165	12.5	175
B	250					215		225
C	400					365		375
A	160	42.5	-	-	-	117.5	15	130
A	200					157.5		170
B	250					207.5		220
C	400					357.5		370



Order:
G104. b1 x h1 x L

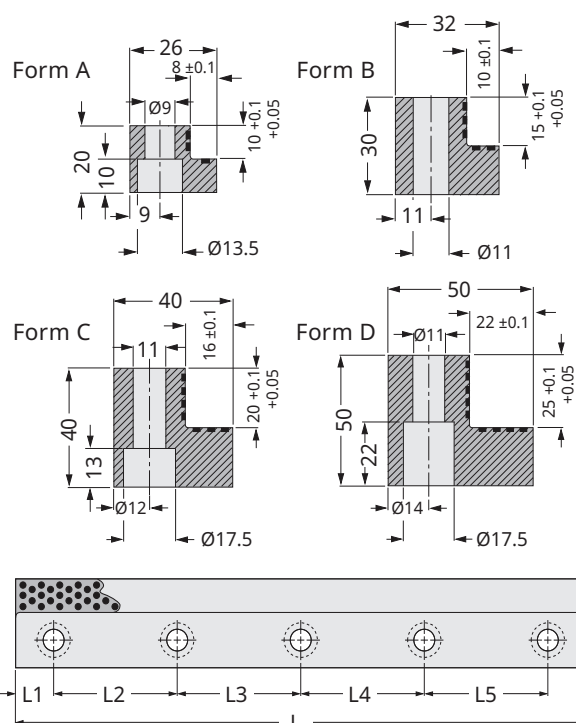
Material:

Bronze + graphite inserts

Wear Plate, "L" Shaped

Code: **G105**

Self-Lubricating & Two-Way Sliding Type



Advantages of Self-Lub. Wear Plate:

- * Highly compatible with sudden movements.
- * Maximum load carrying capacity at low speeds.
- * Can be used under water or with chemical solutions.
- * Wide range of temperature resistance.
- * Resistant to impact in case of vibration.

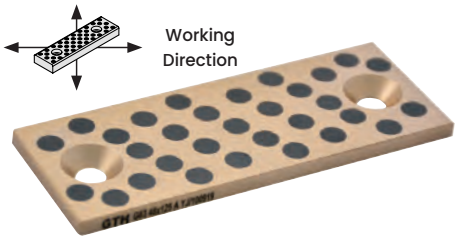
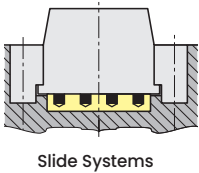
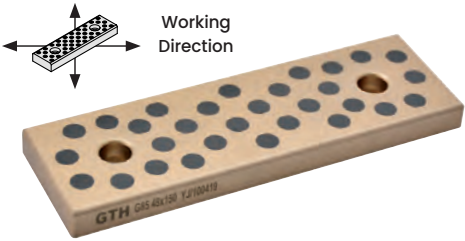
Form	L	L1	L2	L3	L4	L5	Hole
A	100	20	60	-	-	-	2
	150		55	55	-		3
	200		55	50	55		4
B	100	20	60	-	-	-	2
	150		55	55	-		3
	200		55	50	55		4
C	160	15	130	-	-	-	2
	250		80	-	-		3
	200		55	50	55		4
D	250	20	70	70	70	-	4
	300		65	65	65	65	
	350		80	75	75	80	
	350		80	75	75	80	5



Order:
G105. Form x L

Material:

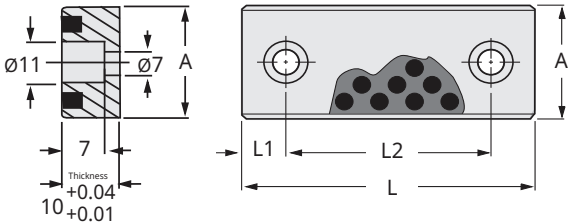
Bronze + graphite inserts



Wear Plate, Self-Lubricating

Code: **G85**

Thickness: 10 mm / with 2 holes



A	L	L1	L2	Hole
18	75	15	45	2 pcs. M6x20
	100	25	50	
	125		75	
	150		100	
28	75	15	45	2 pcs. M6x20
	100	25	50	
	125		75	
	150		100	
38	75	15	45	2 pcs. M6x20
	100	25	50	
	125		75	
	150		100	
48	75	15	45	2 pcs. M6x20
	100	25	50	
	125		75	
	150		100	



Order: **G85**. A x L

Material:
Bronze + Graphite Inserts

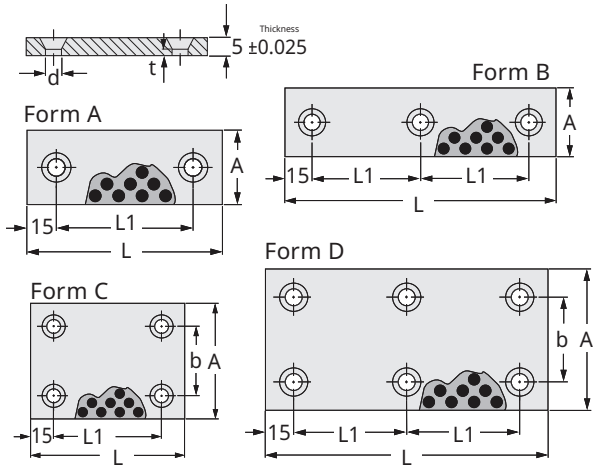


Self-lubricating components provides load carrying capacity beyond expectations in low sliding speeds and a wide temperature range. The graphite inserts are positioned with an appropriate geometric structure. By this means, maximum lubricating effect is achieved during sliding. They especially work better with hardened and ground bearings. Sliding surfaces should be slightly lubricated with lithium grease during mounting before starting to work. 25-30% of the surface in plain, self-lubricating bearings and pillar bearings are formed with graphite inserts. The part corresponding to self-lubricating bearing component should be ground and fixed to the sliding axis in parallel.

Wear Plate, Self-Lubricating

Code: **G83**

Thickness: 5 mm / Form: A - B - C - D



A	L	L1	b	d	t	Form	Screw
18	50	20	-	6.5	1	A	2 pcs. M6x10
	75	45				B	3 pcs.
	100	70					
	150	60				B	3 pcs.
28	50	20	-	9	0.5	A	2 pcs. M8x10
	75	45				B	3 pcs.
	100	70					
	150	60				B	3 pcs.
38	50	20	-	9	0.5	A	2 pcs. M8x16
	75	45				B	3 pcs.
	100	70					
	150	60				B	3 pcs.
48	75	45	-	9	0.5	A	2 pcs. M8x16
	100	70				B	3 pcs.
	125	95					
	150	60				B	3 pcs.
75	75	45	45	9	0.5	C	4 pcs. M8x16
	100	70				D	6 pcs.
	125	95					
	150	60				D	6 pcs.
100	100	70	70	9	0.5	C	4 pcs. M8x16
	125	95				D	6 pcs.
	150	60				D	6 pcs.



Order: **G83**. A x L

Material:
Bronze + Graphite Inserts



Self-lubricating plates are generally preferred because of their strength against lateral loads in large dies or injection moulds. They are maintenance-free, self-lubricating and work long time.

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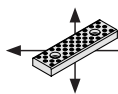
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Guiding Components

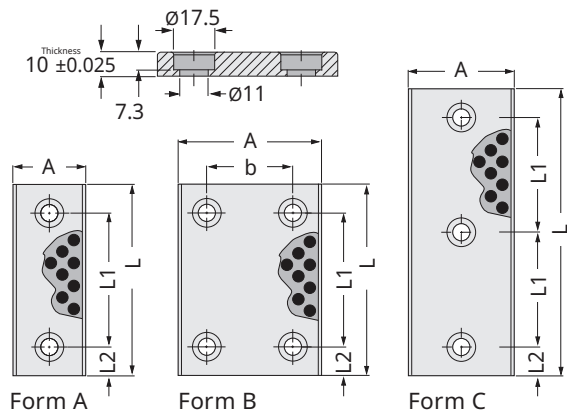


Working
Direction

Wear Plate

Code: **G84**

Self-Lubricating, Thickness: 10 mm / Form: A - B - C



A	L	L1	L2	b	Form
28	75	45	15	-	A
	100	50			
	125	75	25		
	150	100			
38	75	45	15	-	A
	100	50			
	125	75	25		
	150	100			
48	75	45	15	-	A
	100	50			
	125	75	25		
	150	100			
58	75	45	15	-	A
	100	50			
	125	75	25		
	150	100			

A	L	L1	L2	b	Form
58	150		25	-	A
	100		50		
	200				
75	75	45	15	-	A
	100	50			
	125	75	25		
	150	100			
100	100	50		-	B
	125	75			
	150	100	25		
	200	150			
125	150	100		-	B
	200	150	25		
	250	200			
	250	200			
150	150	100		-	B
	200	150	25		
	200	150			



Self-Lubricating
G87

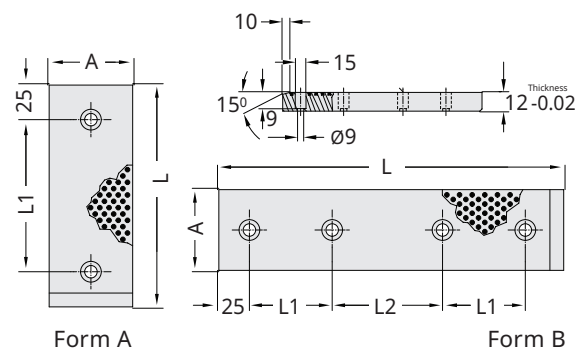


Steel Type
G87C

Wear Plate, VDI 3357

Code: **G87 - G87C**

Self-Lubricating, Thickness: 12 mm / Form: A - B



A	L	L1	L2	Form
30	80	30	-	A
	100	50		
	125	75		
	160	110		
	200	150		
30	250		80	B
	260	60	90	
	280		110	
	300	80	90	
	320		110	
40	80	30	-	A
	100	50		
	125	75		
	160	110		
	200	150		
50	80	30	-	A
	100	50		
	125	75		
	160	110		
	200	150		
50	250	60	80	B
	300	80	90	
	350	100	100	
	400	120	110	
	400	120	110	

A	L	L1	L2	Form
60	80	30	-	A
	100	50		
	125	75		
	160	110		
	200	150		
60	240		70	B
	250	60	80	
	260		90	
	280		110	
	280		110	
80	80	30	-	A
	100	50		
	125	75		
	160	110		
	200	150		
80	240		70	B
	250	60	80	
	260		90	
	280		110	
	280		110	
100	260	60	90	B
	280		110	
	280		110	



Order: **G84. A x L**



* Fixing screws: M10x20



Order:
G87. A x L



for steel type
Order:
G87C. A x L



Self-Lubricating
G91

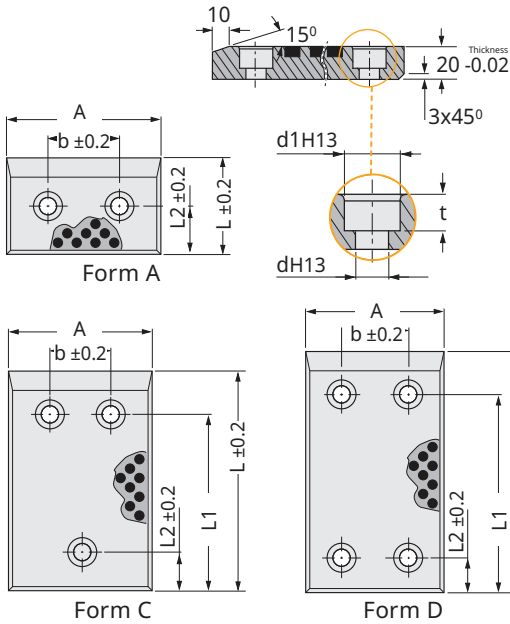


Steel Type
G91C

Wear Plate, VDI 3357

Code: **G91 - G91C**

Self-Lubricating, Thickness: 20 mm / Form: A - C - D



A	L	L1	L2	b	d	d1	t	Screw	Form
80	50	-	25	30	9	15	9	M8x25	A
100	50	-	25	20	13.5	20	13	M12x25	A
	80	-	40						
	80	-	40	75	13.5	20	13	M12x25	A
125	50	-	25					M12x25	A
	80	-	40						
	100	75							
	125	100	25	75	13.5	20	13	M12x25	C
	160	135							
	200	175							
	250	210	40						
160	315	275							
	50	-	25	110	13.5	20	13	M12x25	A
	80	-	40						
	100	75							
	125	100	25						
	160	135		110	13.5	20	13	M12x25	C
	200	175							
160	250	210	40						
	315	275							
	250	210	40	110	13.5	20	13	M12x25	D
	315	275							

for steel type



Order:
G91. A x L



Order:
G91C. A x L



Self-Lubricating
G90

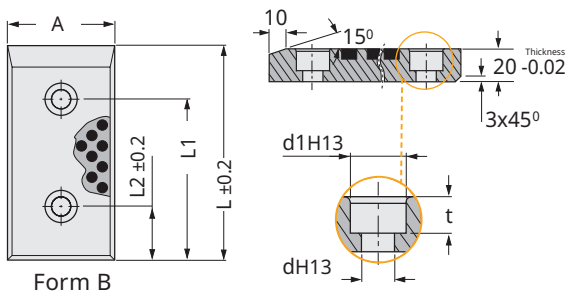


Steel Type
G90C

Wear Plate, VDI 3357

Code: **G90 - G90C**

Self-Lubricating, Thickness: 20 mm



Working Direction
Code: **G91 - G91C / G90 - G90C**

Heat Resistance 150°C
Code: **G91 - G91C / G90 - G90C**

A	L	L1	L2	d	d1	t	Screw	Form
50	80	55		9	15	9	M8x25	B
	100	75						
	125	100	25	13.5	20	13	M12x25	
	160	135						
	200	175						
80	80	55						B
	100	75						
	125	100	25	13.5	20	13	M12x25	
	160	135						
	200	175						
	250	210	40					
	315	275						
100	100	75						B
	125	100	25	13.5	20	13	M12x25	
	160	135						
	200	175						
	250	210	40					
	315	275						
	250	210	40					
	315	275						

for steel type



Order:
G90. A x L



Order:
G90C. A x L

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Guiding Components



NEW

GLOBAL STANDARD COMPONENTS
NAAMS

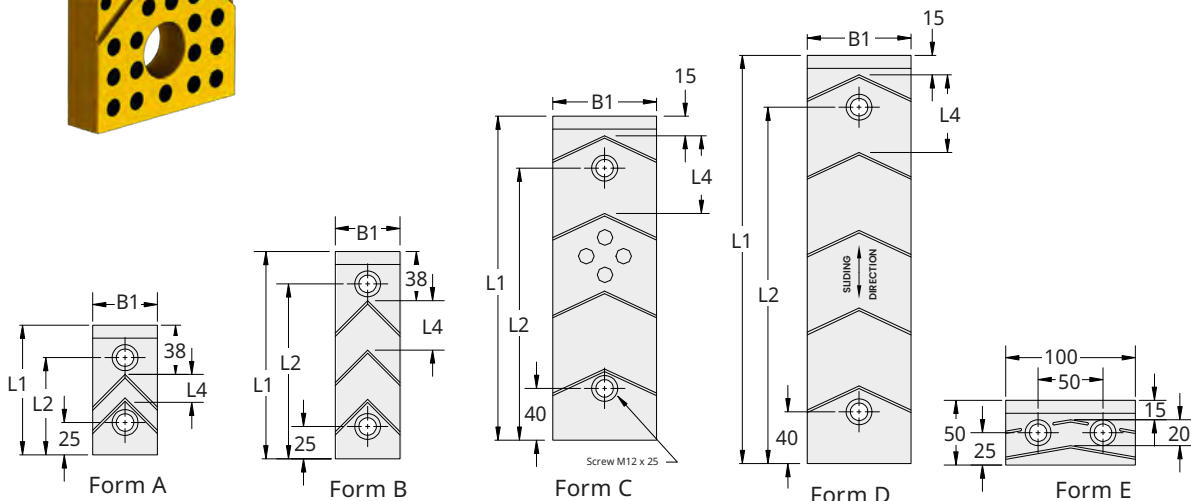


3D CAD
FILES
GÜVENAL

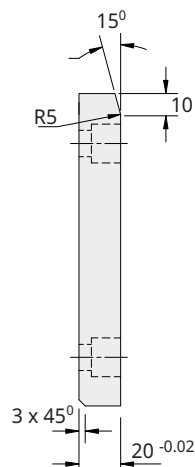
Wear Plate with Visual Wear Indicators, WDX138011

Self-Lubricating, Thickness: 20 mm

Code: **G78**



Order Code	B1	L1 ±0.2	L2 ±0.2	L4	Form
G7850100A	50	100	75	18	A
G7850125A		125	100	38	
G7850160B	50	160	135	38	B
G7850200B		200	175	50	
G7880100A	80	100	75	20	A
G7880125A		125	100	40	
G7880160B	80	160	135	40	B
G7880200B		200	175	60	
G7880250C	80	250	210	60	C
G7880315D	80	315	275	60	D
G7810050E	100	50	-	-	E
G78100100A	100	100	75	20	A
G78100125A		125	100	40	
G78100160B	100	160	135	40	B
G78100200B		200	175	60	
G78100250C	100	250	210	60	C
G78100315D	100	315	275	60	D



Material:
Bronze + Graphite Inserts

Heat Resistance: 150°C



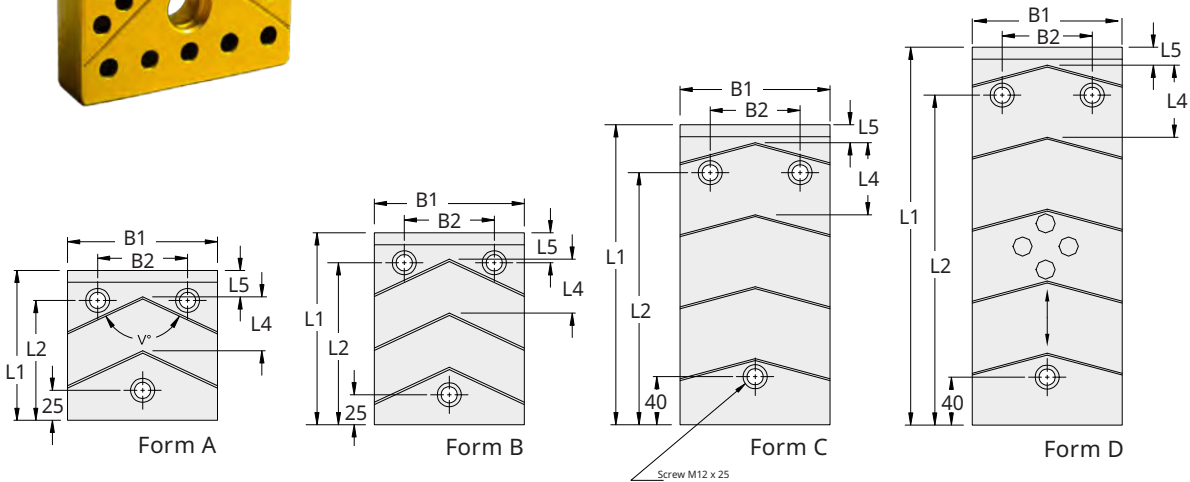
NEW

GLOBAL STANDARD COMPONENTS
NAAMS

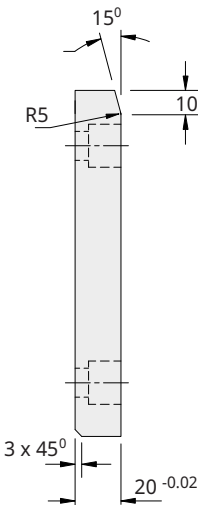


Wear Plate with Visual Wear Indicators, WDX138012
Self-Lubricating, Thickness: 20 mm

Code: **G78-12**



Order Code	B1	B2 ±0.2	L1 ±0.2	L2 ±0.2	L4	L5	Form
G78-12.125100A	125	75	100	75	35	22	A
G78-12.125125A			125	100	45	22	
G78-12.125160B	125	75	160	135	45	22	B
G78-12.125200B			200	175	60	22	
G78-12.125250C	125	75	250	210	60	15	C
G78-12.125315D	125	75	315	275	60	15	D
G78-12.160100A	160	110	100	75	40	15	A
G78-12.160125A			125	100	50	15	
G78-12.160160B	160	110	160	135	50	15	B
G78-12.160200B			200	175	60	15	
G78-12.160250C	160	110	250	210	60	15	C
G78-12.160315D	160	110	315	275	60	15	D

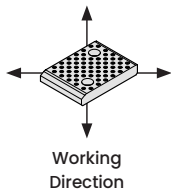


Material:
Bronze + Graphite Inserts

Heat Resistance: 150°C

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Guiding Components

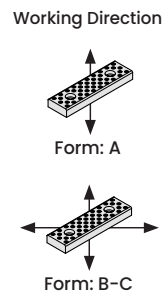
Code: **G89**

A	L	L1	L2	b	Form
50	80	35	20	-	A
	100	55			
	125	80			
	160	115			
	200	155			
	250	100	20	-	C
80	50	-	25	40	D
	80	35	20	40	B
	100	55			
	125	80			
	160	115	20	40	E
	200	155			
	250	100	20	40	F
	315	132			
100	50	-	25	60	D
	80	35	20	60	B
	100	55			
	125	80	20	60	E
	160	115			
	200	155			
	250	100	20	60	F
	315	132			



Order:
G89. A x L

- * Fixing screws: M12 x 25
- * Heat Resistance 150°C.

Code: **G88**

- * Highly compatible with sudden movements.
- * Maximum load carrying capacity at low speeds.
- * Can be used under water or with chemical solutions.
- * Wide range of temperature resistance.
- * Resistant to impact in case of vibration.

A	L	L1	L2	c	b	Form
28	75	45	15	-	14	A
	100	50	25			
	125	75				
	150	100				
38	75	45	15	-	19	A
	100	50	25			
	125	75				
	150	100				
	200	150				
48	75	45	15	-	24	A
	100	50	25			
	125	75				
	150	100				
	200	150				
58	75	45	15	-	29	A
	100	50	25			
	150	100				
75	75	45	15	-	37.5	A
	75	25	25	25	25	B
	100	50	25	-	37.5	A
	100	50	25	25	25	B
	125	75				
	150	100	25	-	37.5	A
	200	150				



Order:
G88. A x L x Form

- * Fixing screws: M10 x 25
- * Heat Resistance 150°C.

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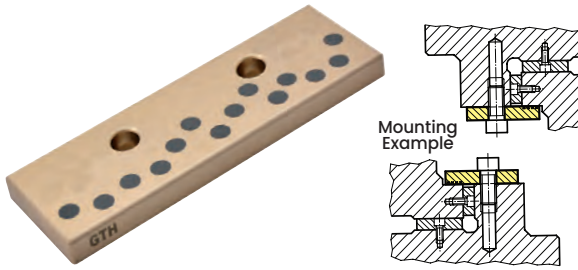
15

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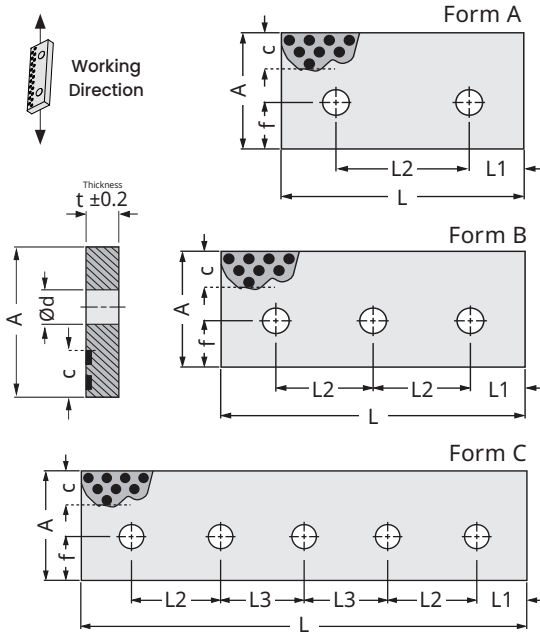
18

Guiding Components



Wear Plate, VDI 3357
Self-Lubricating

Code: **G95**

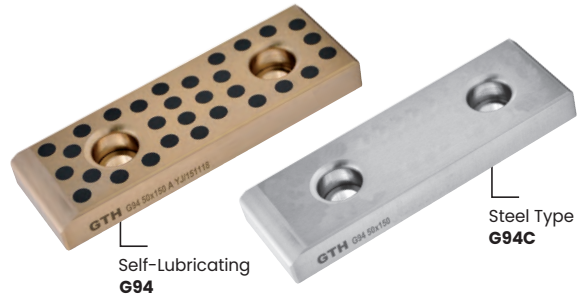


A	L	L1	L2	L3	t	c	f	d	Form				
35	160	45	70	-	10	10	15	11	A				
	200		110						B				
	250		80						B				
45	160	45	70	-	15	15	15	13.5	A				
	200		110						B				
	250		80						B				
55	160	45	70	-	15	20	20	17.5	A				
	200		110						B				
	250		80						B				
75	160	45	70	-	25	25	35	17.5	A				
	200		110						B				
	250		80						B				
100	160	45	70	-	25	30	40	17.5	A				
	200		110						B				
	250		80						C				
	400		80	75	30	30	40	22	A				
	160		70	-					30	30	50	22	B
	200		110										C
	250		80										C
	400		80	75					C				
	125		160	45					70	-	30	30	50
200		110	B										
250		80	C										
400		80	75		C								



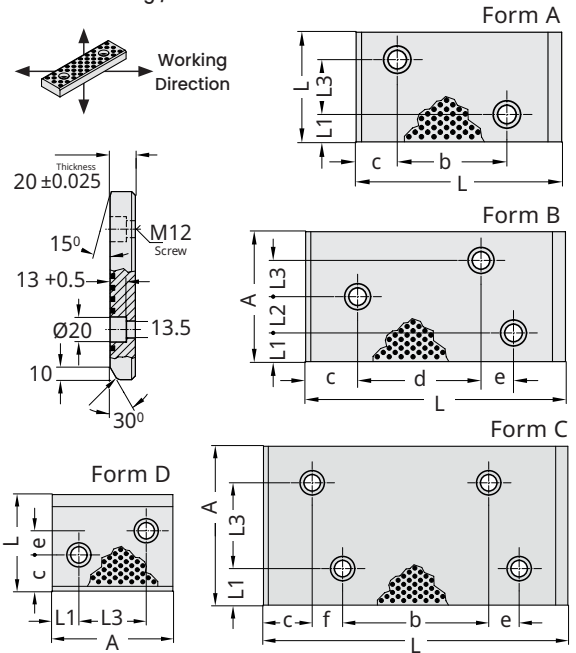
Order:
G95. A x L x t

Material:
Bronze + Graphite Inserts



Wear Plate, NAAMS
Self-Lubricating / Steel

Code: **G94 - G94C**



A	L	L1	L2	L3	c	f	b	e	Form			
50	100	25	-	-	30	-	30	-	A			
	150						80					
	200						120					
80	100	20	-	40	30	-	30	-	A			
	150						80					
	200						120					
	250						170					
	315		210	25	B							
100	50	22	-	56	14	-	-	13	D			
	80				-		20					
	100				30		-	30	-	A		
	150		80	-				A				
	200		28	28				40	-	95	25	B
	250									145		
	315				210							
125	80	25	-	75	30	-	-	20	D			
	100						30	-	A			
	150						80					
	200		37	38	40		-	95	25	B		
	250							145				
	315							-			75	25
160	100	30	-	100	30	-	30	-	A			
	150						80					
	200						95			B		
	250		120	25	C							
	315		-				100		25	185		



Order:
G94. A x L

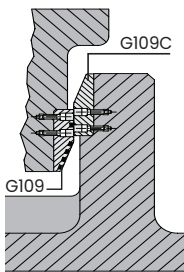


Order:
G94C. A x L

for steel type

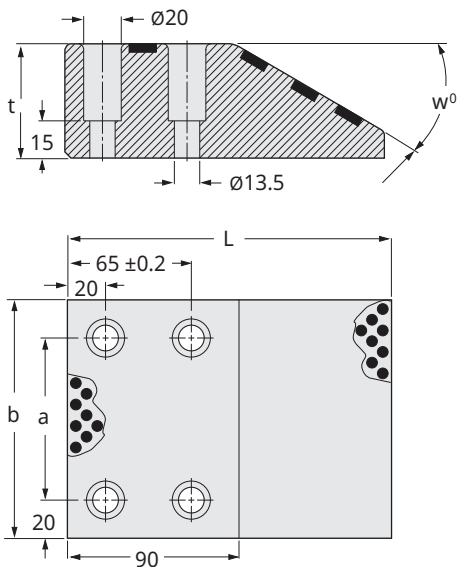


Mounting example



Cam Driver Plate, VDI 3357

Code: **G109**



b	L	t	a	w
100	170	45	60	20°
125			85	
150			110	
100	150	60	60	30°
100	170		60	
125	150		85	
125	170		85	
150	150		110	
150	170		110	



Order:
G109. b x L x t

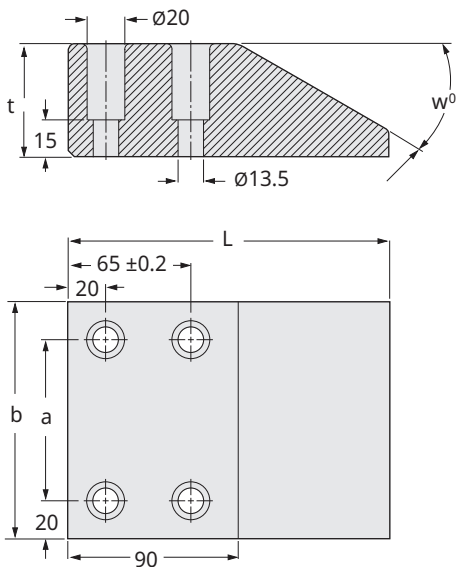
Material:
Bronze + graphite inserts



Self-Lub. Angular Wear Plates for Upper and Lower Body of Cam Units Tolerance between hole distances is ± 0.25 . ± 0.25 is used for all dimensions where tolerance is not specified.

Cam Driver Plate, VDI 3357

Code: **G109C**



b	L	t	a	w
100	170	45	60	20°
125			85	
150			110	
200			160	
100	150	60	60	30°
100	170		60	
125	150		85	
125	170		85	
150	150		110	
150	170		110	
200	150	45	160	30°
200	170			



Order:
G109C. b x L x t

Material:
Steel, Hardened



Steel Type, Angular Wear Plates for Upper and Lower Body of Cam Units Tolerance between hole distances is ± 0.25 . ± 0.25 is used for all dimensions where tolerance is not specified.

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Guiding Components

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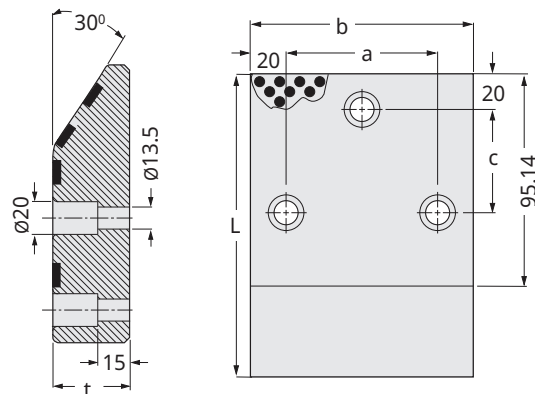
Self-Lubricating
G108



Steel Type
G108C

Cam Driver Plate

Code: **G108 - G108C**



b	L	t	a	c
100	125	30	60	50
	150	45		45
	170	60		45
125	125	30	85	50
	150	45		45
	170	60		45
160	125	30	120	50
	150	45		45
	170	60		45



Order:
G108. b x L

for steel type



Order:
G108C. b x L



Self-Lub. & Steel Angular Wear Plates for Upper and Lower Body of Cam Units
Tolerance between hole distances is ± 0.25 . ± 0.25 is used for all dimensions where tolerance is not specified.

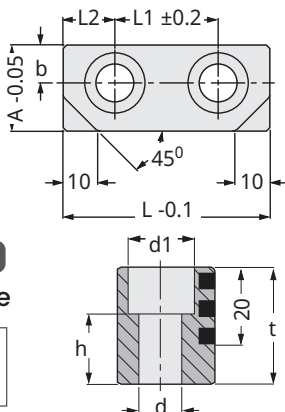


Code: **G76**

Return Cam Plate



Order:
G76. A x L



A	L	b	t	L1	L2	d	d1	h
25	60	11	30	30	15	11	17.5	18
32	60	16	38	30	15	13	20	23
32	80	16	38	40	20	13	20	23



Self-Lubricating
G106

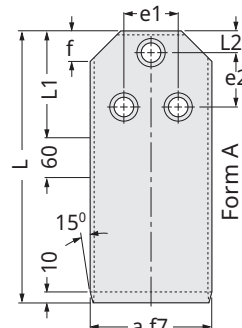
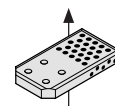
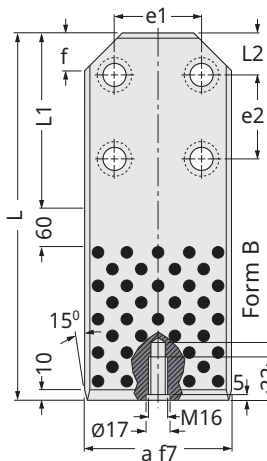
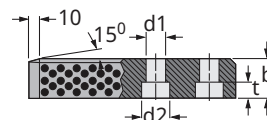


Steel Type
G106C

Guide Bar, VDI 3387

Self-Lubricating & Steel Types

Code: **G106 - G106C**



a	b	L	L1	L2	e1	e2	f	d1	d2	t	Form
63	36	200	90	20	36	50	18	14	20	16	A
71											
90	45		100		50		28	18	26	21	
112		250									
112											
140	45	315	150	90	80	36	22	33	25.5	B	
140											
140	56	400									40
190		500	250								
240											
240	630										

G106

Material:
Bronze + graphite inserts

G106C

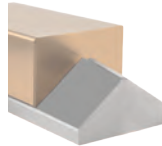
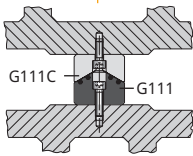
Material:
Steel, Hardened



Order:
G106. a x b x L

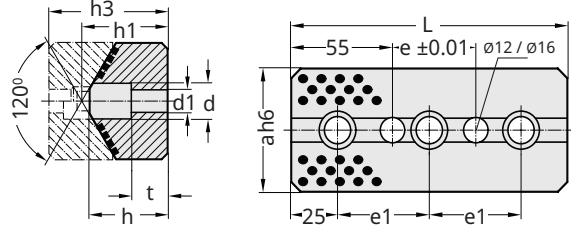


Order:
G106C. a x b x L



G111 Self-Lubricating
Cam Plate, "V" Guide - VDI 3357

Code: **G111**



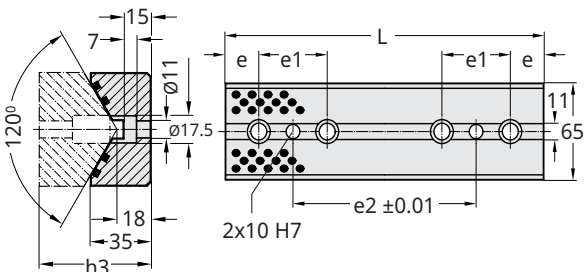
a	L	h1	h	h3	e	e1	d	d1	t	Hole
65	150	47	44	65	45	100	20	13.5	20	2
	200				95	150				3
	250				145	100				
	300				195	125				
125	150	52	47	85	45	100	26	17.5	15	2
	200				95	150				3
	250				145	100				
	300				195	125				

Order:
G111. a x L

There is ± 0.12 tolerance between
fixing holes and ± 0.10 tolerance
between pin holes.

G139 Self-Lubricating
Cam Plate, "V" Guide

Code: **G139**



h3	L	e	e1	e2	Hole
65	100	25	50	20	2
	150			50	3
	200			100	4
	250			150	5
	300			200	6

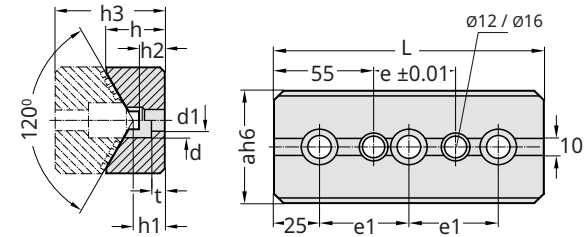
Order:
G139. h3 x L

Material:
Bronze + graphite inserts



G111C Steel
Cam Plate, "V" Guide - VDI 3357

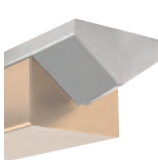
Code: **G111C**



a	L	h1	h	h2	h3	e	e1	d	d1	t	Hole
65	150	18	35	17	65	45	100	20	13.5	8	2
	200					95	150				3
	250					145	100				
	300					195	125				
125	150	33	60	32	85	45	100	26	17.5	15	2
	200					95	150				3
	250					145	100				
	300					195	125				

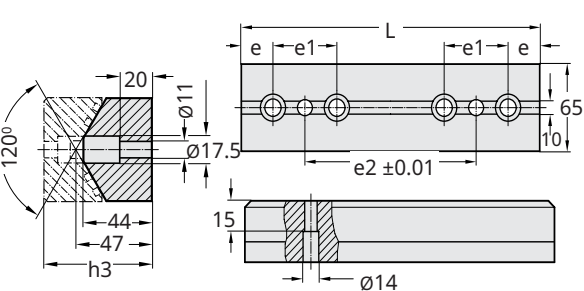
Order:
G111C. a x L

Material:
Steel, Hardened



G139C Steel
Cam Plate, "V" Guide

Code: **G139C**



h3	L	e	e1	e2	Hole
65	100	25	50	20	2
	150			50	3
	200			100	4
	250			150	5
	300			200	6

Order:
G139C. h3 x L

Material:
Steel, Hardened

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Guiding Components